

Software Requirements Specification

Octagon AI — UFC & MMA Analytics Platform

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1. Executive Summary

Octagon AI is transitioning from a static frontend prototype into a fully functional, production-ready SaaS platform. The system will leverage automated data scraping to provide live UFC/MMA analytics, dynamic Elo rankings, AI-driven prediction models, and an integrated Stripe paywall for premium subscriber access.

2. Technology Stack & Architecture

The application will be built using a robust, scalable modern web stack optimized for performance and rapid data processing.

- Frontend** **Next.js (React)** - For server-side rendering, SEO optimization, and dynamic UI components.
- Backend** **Node.js / Next.js API Routes** - Handling logic, cron jobs, and secure API endpoints.
- Database** **Neon DB (PostgreSQL)** - A scalable, serverless relational database for storing fighters, events, and user data securely.

- **Hosting** **Vercel** - For continuous integration and high-performance edge deployment.
- **Payments** **Stripe** - Secure processing of monthly subscriptions.

3. Core System Features

3.1 Authentication & User Management

- **Secure Onboarding:** Full signup, login, and logout flow using encrypted password handling.
- **User Profiles:** Personalized dashboards where users can view their subscription status and manage saved data.
- **UI Preferences:** Persistent Dark/Light mode toggling to match user system preferences.

3.2 Automated Data Pipelines (Cron Jobs)

The platform relies heavily on 100% accurate, automated data. The backend will routinely fetch, sanitize, and insert data from external sources.

- **Sources:** `roster.watch`, `Topology`, `ESPN`, `UFC`, and `FanDuel`.
- **Live Odds Sync:** Automatically draw live fight odds from FanDuel, factoring into the prediction engine to find and highlight betting edges.
- **Roster Sync:** Auto-import the full active roster, continuously cross-referencing to add new signings and remove duplicates.
- **Event Sync:** Automatically pull fight cards, bout orders, and scheduled dates.
- **Post-Event Trigger:** Following an event, the system will auto-archive the event, update fight results, recalculate Elo rankings, and update finish stats.

3.3 Comprehensive Fighter Database

Each fighter entity will house deep statistical data, updated automatically:

- **Biometrics:** Age, Weight Class, Division, Height, and Reach.
- **Record Details:** Wins, Losses, Draws, and detailed finish statistics (KO/TKO, Sub).
- **Advanced Analytics:** Performance metrics (SLpM, striking accuracy, takedown defense) and proprietary Elo Ratings.
- **AI Predictions:** Model-driven statistics for upcoming matchups.

4. Paywall & Subscription Tiers

Free Tier (Unregistered / Non-paying Users)

- Homepage: Only the **most upcoming UFC event** is fully viewable.
- **Events Tab:** Viewable (Prediction picks remain locked).
- **Elo Rankings & Fighters Tabs:** Fully accessible.
- **Model & Performance Tabs:** Fully accessible.

Premium Tier (Monthly Paid Subscribers)

- **Homepage Unlocked:** Full access to all homepage content and historical event data.
- **Betting Tab:** Unlocked access to premium betting insights and odds.
- **Matchup Lab:** Unlocked access to the interactive comparison tool.
- **Prediction Picks:** Full visibility of AI-driven prediction picks across all upcoming fights.

Client Approval & Scope Freeze

This document details the complete functional scope of the Octagon AI platform. **Please review carefully.** Any new features, logic changes, or additional data sources requested after development begins will be considered a scope change, which may require budget adjustments and timeline extensions. If everything looks correct, your approval will initiate the coding phase.